7D_3 is the Case where the curled geometry is a 3-sphere, S^3.

Notation and Setup:

Starting Metric for this Case: $g_{00}=\frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}$, $g_{11}=-a[t]^2\,B[r]$, $g_{22}=-r^2\,a[t]^2\,C[r]$, $g_{33}=-r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]$, $g_{44}=-a[t]^2\,\mathcal{D}_4[\Psi]$, $g_{55}=-\Psi^2\,a[t]^2\,\mathcal{D}_5[\Psi]$, $g_{66}=-\Psi^2\,a[t]^2\,\text{Sin}[\psi_5]^2\,\mathcal{D}_5[\Psi]$, a(t) not constant

Corresponding Line Element

$$ds^2 = \frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}\,dt^2 - a[t]^2\,B[r]\,dr^2 - r^2\,a[t]^2\,C[r]\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]\,d\phi^2 - a[t]^2\,\mathcal{D}_4[\Psi]\,d\Psi^2 - \Psi^2\,a[t]^2\,\mathcal{D}_5[\Psi]\,d\psi_5^2 - \Psi^2\,a[t]^2\,\text{Sin}[\psi_5]^2\,\mathcal{D}_5[\Psi]\,d\psi_6^2$$

Output Display Simplification - Parameters are sometimes suppressed
Prime (') means derivative wrt r.
Dot (.) means derivative wrt t.
Examples: **A = A(t,r)**, **$A'=\partial_r A(t,r)$** and **$\dot{A}=\partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: The exact solution to the Λ vacuum field equations that has (1) C = 1, (2) a(t) is a free parameter, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the a(t) formula).

FG-GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameters a and C, and (2) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the a(t) formula).

a: Free-gauge GRW Solution

Until now, the only case that was simple enough to directly solve for FG-GRW (followed by GRW as a special case) was 4D. However, the curled space in 7D_3 is a 3-sphere, and the solution mimics 4D. We find that we can again solve for FG-GRW first.

The starting metric and line element above are the result of the following assumptions

4-space and curled space assumptions.

- 4-space:

$A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2}$, the standard Free-gauge GRW formula for 7D,

$B(r)$, a free parameter whose formula we seek

$C(r)$ is a free parameter (meant to stay free)

- Curled-space:

  A compact 3-sphere having the areal-radial metric.

  The areal-radial metric is derived from the 'Round Metric':

  (1) $ds^2 = p_4^2\, d\psi_4^2 + p_4^2 \sin^2\psi_4\, (d\overline{\theta}^2 + \sin^2\overline{\theta}\, d\overline{\phi}^2)$

  where p_4 is the 3-sphere's radius and $\overline{\theta}$ and $\overline{\phi}$ are its angles.

  Substitution to areal-radial form:

  (2) $\Psi = p_4 \sin\psi_4$ where Ψ is the areal radius

$\Rightarrow \quad 0 \le \Psi < p_4$.

Observe that as Ψ moves from radius 0 to radius p_4, we can imagine circles on earth (a 2-sphere S^2, one dimension down from this case) that begin as a point at the north pole and become larger and larger latitude circles until they reach the equator. The circles have covered only half the globe. This illustrates why the areal-radius chart covers only half of the S^3.

(3) $d\Psi \overset{(2)}{=} p_4 \cos\psi_4\, d\psi_4$, or $d\psi_4 = \frac{d\Psi}{p_4 \cos\psi_4}$

(4) $\cos^2\psi_4 = 1 - \sin^2\psi_4 \overset{(2)}{=} 1 - \frac{\Psi^2}{p_4^2}$

(5) $p_4^2\, d\psi_4^2 \overset{(3)}{=} \frac{p_4^2\, d\Psi^2}{p_4^2 \cos^2\psi_4} \overset{(4)}{=} \frac{d\Psi^2}{1 - \frac{\Psi^2}{p_4^2}}$

(6) $p_4^2 \sin^2\psi_4 \overset{(2)}{=} \Psi^2$

(7) $ds^2 \overset{(1,5,6)}{=} \frac{d\Psi^2}{1 - \frac{\Psi^2}{p_4^2}} + \Psi^2(d\overline{\theta}^2 + \sin^2\overline{\theta}\, d\overline{\phi}^2)$

Relabel $\overline{\theta} \to \psi_5$ and $\overline{\phi} \to \psi_6$, and apply scale factor $a(t)$

and free parameters $\hat{D}_4$ and D_5:

(8) $ds^2 = a[t]^2 \{ \frac{\hat{D}_4[\Psi]}{1 - \frac{\Psi^2}{p_4^2}} d\Psi^2 + D_5[\Psi]\Psi^2(d\psi_5^2 + \sin^2\psi_5\, d\psi_6^2) \}$

Note: we have continued to use lower-case Greek to denote angles (ψ_4 - ψ_6) and we use upper-case Ψ to represent the radius.

Define free parameter $\mathbb{D}_4$:

$$(9)\quad \mathbb{D}_4(\Psi) = \frac{\hat{\mathbb{D}}_4[\Psi]}{1 - \frac{\Psi^2}{p_4{}^2}},$$

The resulting compact sphere line element is:

$$(10)\quad ds^2 = a[t]^2\{\ \mathbb{D}_4[\Psi]d\Psi^2 + \mathbb{D}_5[\Psi]\Psi^2(d\psi_5{}^2 + \sin^2\psi_5\ d\psi_6{}^2)\ \}$$

Compare the standard 3-space line element to the S^3 line element (10):

$$ds^2 = a[t]^2\{\ B[r]dr^2 + C[r]r^2(d\Theta^2 + \sin^2\Theta\ d\phi^2)\ \}$$

- $\mathbb{D}_4(\Psi)$ corresponds to $B(r)$,
- $\mathbb{D}_5(\Psi)$ corresponds to $C(r)$, and
- Ψ corresponds to r.

In all prior cases, C has been the sole free parameter in the Free-gauge GRW formula. In Case 7D_3, $\mathbb{D}_5$ is a second free parameter.

Summarizing, our curled space assumptions are:

- $\mathbb{D}_4(\Psi)$ is a free parameter whose formula we seek,
- $\mathbb{D}_5(\Psi)$ is also a free parameter, and
- $0 \le \Psi < p_4$.

As an aside, notice that we have not introduced a $\mathbb{D}_6$ parameter.

The initial field equations have 3 unknowns, B, $\mathbb{D}_4$, and $\mathbb{D}_5$:

R_{00}: $\Lambda = \Lambda$

$$R_{11}:\quad -\frac{5k}{a^2} + \Lambda - \frac{2C'}{a^2 r C B} + \frac{(C')^2}{2a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C'B'}{2a^2 C B^2} - \frac{C''}{a^2 C B} = \Lambda$$

$$R_{22}:\quad -\frac{5k}{a^2} + \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \Lambda - \frac{2C'}{a^2 r C B} + \frac{B'}{2a^2 r B^2} + \frac{C'B'}{4a^2 C B^2} - \frac{C''}{2a^2 C B} = \Lambda$$

$$R_{33}:\quad -\frac{5k}{a^2} + \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \Lambda - \frac{2C'}{a^2 r C B} + \frac{B'}{2a^2 r B^2} + \frac{C'B'}{4a^2 C B^2} - \frac{C''}{2a^2 C B} = \Lambda$$

$$R_{44}:\quad -\frac{5k}{a^2} + \Lambda + \frac{\mathbb{D}_4'[\Psi]}{a^2 \Psi \mathbb{D}_4^2} - \frac{2\mathbb{D}_5'[\Psi]}{a^2 \Psi \mathbb{D}_4 \mathbb{D}_5} + \frac{\mathbb{D}_4'[\Psi]\mathbb{D}_5'[\Psi]}{2a^2 \mathbb{D}_4^2 \mathbb{D}_5} + \frac{\mathbb{D}_5'[\Psi]^2}{2a^2 \mathbb{D}_4 \mathbb{D}_5^2} - \frac{\mathbb{D}_5''[\Psi]}{a^2 \mathbb{D}_4 \mathbb{D}_5} = \Lambda$$

$$R_{55}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{1}{a^2\,\Psi^2\,\mathcal{D}_4}+\frac{1}{a^2\,\Psi^2\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]}{2\,a^2\,\Psi\,\mathcal{D}_4^2}-\frac{2\,\mathcal{D}_5'[\Psi]}{a^2\,\Psi\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]\,\mathcal{D}_5'[\Psi]}{4\,a^2\,\mathcal{D}_4^2\,\mathcal{D}_5}-\frac{\mathcal{D}_5''[\Psi]}{2\,a^2\,\mathcal{D}_4\,\mathcal{D}_5}=\Lambda$$

$$R_{66}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{1}{a^2\,\Psi^2\,\mathcal{D}_4}+\frac{1}{a^2\,\Psi^2\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]}{2\,a^2\,\Psi\,\mathcal{D}_4^2}-\frac{2\,\mathcal{D}_5'[\Psi]}{a^2\,\Psi\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]\,\mathcal{D}_5'[\Psi]}{4\,a^2\,\mathcal{D}_4^2\,\mathcal{D}_5}-\frac{\mathcal{D}_5''[\Psi]}{2\,a^2\,\mathcal{D}_4\,\mathcal{D}_5}=\Lambda$$

R_{33} and R_{66} are redundant, the same as R_{22} and R_{55}, respectively.

R_{00} is an identity and the off-diagonal R_{01} equation
does not appear because we have already set $A(t,r)=\mathcal{A}(t)$.

$R_{11}-2R_{22}$ can be algebraically solved for B:

$$R_{11}-2R_{22}: \quad \frac{5\,k}{a^2}-\frac{2}{a^2\,r^2\,C}+\frac{2}{a^2\,r^2\,B}-\Lambda+\frac{2\,C'}{a^2\,r\,C\,B}+\frac{(C')^2}{2\,a^2\,C^2\,B}=-\Lambda$$

$$B(r)=\frac{(2\,C[r]+r\,C'[r])^2}{4\,C[r]\left(1-\frac{5}{2}\,k\,r^2\,C[r]\right)}$$

The remaining equations become:

$$R_{11}: \quad \Lambda=\Lambda$$

$$R_{22}: \quad \Lambda=\Lambda$$

$$R_{44}: \quad -\frac{5\,k}{a^2}+\Lambda+\frac{\mathcal{D}_4'[\Psi]}{a^2\,\Psi\,\mathcal{D}_4^2}-\frac{2\,\mathcal{D}_5'[\Psi]}{a^2\,\Psi\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]\,\mathcal{D}_5'[\Psi]}{2\,a^2\,\mathcal{D}_4^2\,\mathcal{D}_5}+\frac{\mathcal{D}_5'[\Psi]^2}{2\,a^2\,\mathcal{D}_4\,\mathcal{D}_5^2}-\frac{\mathcal{D}_5''[\Psi]}{a^2\,\mathcal{D}_4\,\mathcal{D}_5}=\Lambda$$

$$R_{55}: \quad -\frac{5\,k}{a^2}+\Lambda-\frac{1}{a^2\,\Psi^2\,\mathcal{D}_4}+\frac{1}{a^2\,\Psi^2\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]}{2\,a^2\,\Psi\,\mathcal{D}_4^2}-\frac{2\,\mathcal{D}_5'[\Psi]}{a^2\,\Psi\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{\mathcal{D}_4'[\Psi]\,\mathcal{D}_5'[\Psi]}{4\,a^2\,\mathcal{D}_4^2\,\mathcal{D}_5}-\frac{\mathcal{D}_5''[\Psi]}{2\,a^2\,\mathcal{D}_4\,\mathcal{D}_5}=\Lambda$$

R_{11} and R_{22} are now identities, and $R_{44}-2R_{55}$ can be algebraically solved for $\mathcal{D}_4$:

$$R_{44}-2R_{55}: \quad \frac{5\,k}{a^2}-\Lambda+\frac{2}{a^2\,\Psi^2\,\mathcal{D}_4}-\frac{2}{a^2\,\Psi^2\,\mathcal{D}_5}+\frac{2\,\mathcal{D}_5'[\Psi]}{a^2\,\Psi\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{\mathcal{D}_5'[\Psi]^2}{2\,a^2\,\mathcal{D}_4\,\mathcal{D}_5^2}=-\Lambda$$

$$\mathcal{D}_4[\Psi]=\frac{(2\,\mathcal{D}_5[\Psi]+\Psi\,\mathcal{D}_5'[\Psi])^2}{4\,\mathcal{D}_5[\Psi]\left(1-\frac{5}{2}\,k\,\Psi^2\,\mathcal{D}_5[\Psi]\right)} \text{ where } \mathcal{D}_5[\Psi] \text{ is a free parameter.}$$

Observe the similarity between $B(r)$ and $\mathcal{D}_4[\Psi]$. This suggests that C
and $\mathcal{D}_5$ both be given similar treatment when going between GRW and FG-GRW:

- C: $r \leftrightarrow r\sqrt{C}$ and $1 \leftrightarrow C$
- $\mathcal{D}_5$: $\Psi \leftrightarrow \Psi\sqrt{\mathcal{D}_5}$ and $1 \leftrightarrow \mathcal{D}_5$

Plugging $\mathcal{D}_4$ in, the remaining equations are solved:

$$R_{44}: \quad \Lambda=\Lambda$$

$$R_{55}: \quad \Lambda=\Lambda$$

We have found a 7D_3 Free-gauge GRW metric:

$A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6k+\Lambda\,a[t]^2}$, where curvature k is fixed by the Section c compact-sphere constraint,

$B(r) = \frac{(2C+rC')^2}{4C\left(1-\frac{5}{2}kr^2C\right)}$,

$C(r)$ is a free parameter,

$\mathcal{D}_4(\Psi) = \frac{(2\mathcal{D}_5+\Psi\mathcal{D}_5')^2}{4\mathcal{D}_5\left(1-\frac{5}{2}k\Psi^2\mathcal{D}_5\right)}$,

$\mathcal{D}_5(\Psi)$ is a free parameter,

and line element

$$ds^2 = \frac{6\,\dot{a}[t]^2}{-6k+\Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2\,(2C[r]+rC'[r])^2}{4C[r]\left(1-\frac{5}{2}kr^2C[r]\right)}\,dr^2 - r^2\,a[t]^2\,C[r]\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]\,d\phi^2 - \frac{a[t]^2\,(2\mathcal{D}_5[\Psi]+\Psi\mathcal{D}_5'[\Psi])^2}{4\mathcal{D}_5[\Psi]\left(1-\frac{5}{2}k\Psi^2\mathcal{D}_5[\Psi]\right)}\,d\Psi^2 - \Psi^2\,a[t]^2\,\mathcal{D}_5[\Psi]\,d\psi_5^2 - \Psi^2\,a[t]^2\,\text{Sin}[\psi_5]^2\,\mathcal{D}_5[\Psi]\,d\psi_6^2$$

b: Solve $\mathcal{A}(t) = c^2$ to find a(t)

Steps for finding a(t) by solving $\frac{6\,\dot{a}[t]^2}{-6k+\Lambda\,a[t]^2} = c^2$ were provided in Case 4D_a.

$$a(t) = \begin{cases} \sqrt{6}\,\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\cosh\left(\frac{c\sqrt{\Lambda}\,t}{\sqrt{6}}\right) & \text{if } k>0 \;\&\; \Lambda\neq 0 \\ \sqrt{6}\,\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\sinh\left(\frac{c\sqrt{\Lambda}\,t}{\sqrt{6}}\right) & \text{if } k<0 \;\&\; \Lambda\neq 0 \\ c\,t\,\sqrt{|k|} & \text{if } k<0 \;\&\; \Lambda=0 \\ \alpha\,e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{6}}} & \text{if } k=0 \;\&\; \Lambda\geq 0 \;\&\; \alpha>0 \text{ is a free constant} \end{cases}$$

c: GRW Solution

As mentioned in Section a, we obtain the 7D_3 GRW from FG-GRW by setting $C(r) = \mathcal{D}_5(\Psi) = 1$:

$\mathcal{A}(t) = c^2$ where $k = \frac{2}{5\,p_4^2}$ (see Compact-sphere constraint, below),

$B(r) = \frac{1}{1-\frac{5kr^2}{2}}$,

$C(r) = 1$,

$\mathcal{D}_4(\Psi) = \frac{1}{1-\frac{5k\Psi^2}{2}}$,

$\mathcal{D}_5(\Psi) = 1$,

and line element

$$ds^2 = c^2\, dt^2 - \frac{a[t]^2}{1-\frac{5\,k\,r^2}{2}}\, dr^2 - r^2\, a[t]^2\, d\Theta^2 - r^2\, a[t]^2\, \text{Sin}[\Theta]^2\, d\phi^2 - \frac{a[t]^2}{1-\frac{5\,k\,\Psi^2}{2}}\, d\Psi^2 - \Psi^2\, a[t]^2\, d\psi_5^2 - \Psi^2\, a[t]^2\, \text{Sin}[\psi_5]^2\, d\psi_6^2$$

Compact-sphere constraint:

We show that there is a constraint on k due to the curled space being a sphere of radius p_4:

- The 3-space (r,Θ,ϕ) base line element is

  $$ds^2 = B(r)\, dr^2 + r^2(d\Theta^2 + \text{Sin}^2[\Theta]\, d\phi^2)$$

  where

  $$B(r) = \frac{1}{1-\frac{5}{2}\,k r^2}.$$

- The 3-sphere (Ψ,ψ_5,ψ_6) transverse line element is:

  $$ds^2 = \mathcal{D}_4[\Psi]\, d\Psi^2 + \Psi^2(d\psi_5{}^2 + \text{Sin}^2[\psi_5]\, d\psi_6{}^2)$$

  where

  $$\mathcal{D}_4(\Psi) = \frac{1}{1-\frac{5}{2}\,k\Psi^2}.$$

- A round 3-sphere of radius p_4 was shown in equation (7) to have line element:

(7) $$ds^2 = \frac{d\Psi^2}{1-\frac{\Psi^2}{p_4{}^2}} + \Psi^2(d\psi_5{}^2 + \sin^2\psi_5\, d\psi_6{}^2)$$

$$\Rightarrow \quad -\frac{5}{2}k\Psi^2 = \frac{\Psi^2}{p_4{}^2}$$

or $$\mathbf{k = \frac{2}{5\,p_4{}^2}}.$$

$$\Rightarrow \quad \mathcal{D}_4(\Psi) = \frac{1}{1-\frac{\Psi^2}{p_4{}^2}}$$

and $$B(r) = \frac{1}{1-\frac{r^2}{p_4{}^2}}.$$

So, both 3-space (r,Θ,ϕ) and 3-sphere (Ψ,ψ_5,ψ_6) are spheres of radius p_4.

Radius p_4 is assumed to be a very tiny number, certainly not the radius of our universe. Thus, this case does not describe our universe.

This k-constraint is reflected in the Table in Section e.

d: Kaluza-Klein (KK) bubble-of-nothing

This case does not have a KK bubble constraint - there is no GRW inverse relationship between B and either compact space parameter, D_4 or D_5.

e: FG-GRW Summary Table

The table below shows the FG-GRW solution obtained in Section a and the GRW solution obtained in Section c. It also provides comparison to the previous 7D cases.

		7D_1	7D_2 Solution 1	7D_2 Solution 2	7D_3
GRW	Con-straints	$\Lambda>0,\ k=0$ $\chi=-\zeta_4-\zeta_5-\zeta_6\neq 0$ $\zeta_4=-\frac{\zeta_5\,\zeta_6}{\zeta_5+\zeta_6}$	$k=\frac{1}{5\,p_4^2}>0$	$p_4=1$	$k=\frac{2}{5\,p_4^2}>0$
	KK-bubble Con-straints (optional)	$\zeta_4=\zeta_5=0$ $\chi=-\frac{p_6}{2}$	χ^*	χ^* $k\geq-\frac{1}{5\,p_6^2}$	None
	$\mathcal{A}$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$
	B	$\frac{1}{\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}$	$\frac{1}{\mathcal{D}_6}$	$\frac{1}{\mathcal{D}_6}$	$\frac{1}{1-\frac{5\,k\,r^2}{2}}$
	C	1	1	1	1
	$\mathcal{D}_4$	$\left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_4}{\chi}}$	1	r^2	$\frac{1}{1-\frac{5\,k\,\Psi^2}{2}}$
	$\mathcal{D}_5$	$\left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_5}{\chi}}$			1
	$\mathcal{D}_6$	$\left(1+\frac{\chi}{r}\right)^{-\frac{\zeta_6}{\chi}}$	$1-\frac{5\,k\,r^2}{3}+\frac{\chi}{r}$	$\frac{1}{3}-k\,r^2+\frac{\chi}{r^3}$	
FG GRW	$\mathcal{A}$	$\frac{6\,\dot{a}^2}{-6\,k+a^2\,\Lambda}$	$\frac{6\,\dot{a}^2}{-6\,k+a^2\,\Lambda}$	$\frac{6\,\dot{a}^2}{-6\,k+a^2\,\Lambda}$	$\frac{6\,\dot{a}^2}{-6\,k+a^2\,\Lambda}$
	B	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4\,\mathcal{D}_5\,\mathcal{D}_6}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_6}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_6}$	$\frac{(2\,C+r\,C')^2}{4\,C\left(1-\frac{5}{2}\,k\,r^2\,C\right)}$
	C	C (free)	C (free)	C (free)	C (free)
	$\mathcal{D}_4$	$\left(1+\frac{\chi}{r\,\sqrt{C}}\right)^{-\frac{\zeta_4}{\chi}}$	1	$r^2\,C$	$\frac{(2\,\mathcal{D}_5+\Psi\,\mathcal{D}_5')^2}{4\,\mathcal{D}_5\left(1-\frac{5}{2}\,k\,\Psi^2\,\mathcal{D}_5\right)}$
	$\mathcal{D}_5$	$\left(1+\frac{\chi}{r\,\sqrt{C}}\right)^{-\frac{\zeta_5}{\chi}}$			$\mathcal{D}_5$ (free)
	$\mathcal{D}_6$	$\left(1+\frac{\chi}{r\,\sqrt{C}}\right)^{-\frac{\zeta_6}{\chi}}$	$1-\frac{5}{3}\,k\,r^2\,C+\frac{\chi}{r\,\sqrt{C}}$	$\frac{1}{3}-k\,r^2\,C+\frac{\chi}{r^3\,C^{3/2}}$	

* χ depends on k and p_6 in a slightly lengthy formula

Notes:

- $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation
  - ✓ The solution to the equation is the a(t) formula from Section b
- The two constraint rows display restrictions on parameter constants
  - ✓ Due to the field equations
  - ✓ Due to KK-bubble smoothness

- By contrast, domain restrictions on r (RCPC and its analogs) are not displayed as constraints
- FG-GRW has the same constraints (both rows) as GRW

f: The Vacuum Friedmann Equations

Plugging $\mathcal{A}$, B, C, D_4, and D_5 (but not a) from the GRW solution into the field equations yields:

$$R_{00}: \quad \frac{6\,\ddot{a}}{a\,c^2} = \Lambda$$

$$R_{11}: \quad \frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} - \frac{2\,C'}{a^2\,r\,C} + \frac{15\,k\,r\,C'}{2\,a^2\,C} + \frac{(C')^2}{2\,a^2\,C^2} - \frac{5\,k\,r^2\,(C')^2}{4\,a^2\,C^2} - \frac{C''}{a^2\,C} + \frac{5\,k\,r^2\,C''}{2\,a^2\,C} = \Lambda$$

$$R_{22}: \quad \frac{5\,k}{a^2} - \frac{1}{a^2\,r^2} + \frac{1}{a^2\,r^2\,C} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} - \frac{2\,C'}{a^2\,r\,C} + \frac{25\,k\,r\,C'}{4\,a^2\,C} - \frac{C''}{2\,a^2\,C} + \frac{5\,k\,r^2\,C''}{4\,a^2\,C} = \Lambda$$

R_{22} is redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$$-\frac{1}{30}R_{00} + \frac{1}{5}R_{11}: \quad \frac{k}{a^2} + \frac{\dot{a}^2}{a^2\,c^2} - \frac{2\,C'}{5\,a^2\,r\,C} + \frac{3\,k\,r\,C'}{2\,a^2\,C} + \frac{(C')^2}{10\,a^2\,C^2} - \frac{k\,r^2\,(C')^2}{4\,a^2\,C^2} - \frac{C''}{5\,a^2\,C} + \frac{k\,r^2\,C''}{2\,a^2\,C} = \frac{\Lambda}{6}$$

$$\frac{1}{6}R_{00}: \quad \frac{\ddot{a}}{a\,c^2} = \frac{\Lambda}{6}$$